

Collaborative Multi-Agent Proximal Policy Based Resource Allocation in Clustered IoV Networks

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Abstract—The Internet of Vehicles (IoV) necessitates efficient resource management to meet the growing demands for high data rates, low latency, and real-time communication in Intelligent Transportation Systems (ITS). This paper presents a novel multi-agent reinforcement learning framework based on the Proximal Policy Optimization (PPO) algorithm for optimizing resource allocation in clusters of cells within IoV networks. The framework dynamically allocates transmission power and bandwidth across base stations, each acting as a collaborative agent. Extensive simulations demonstrate that the multi-agent PPO approach outperforms other reinforcement learning algorithms, including Soft Actor-Critic (SAC) and Advantage Actor-Critic (A2C), regarding throughput, power efficiency, dynamic bandwidth allocation, and overall network performance.

Index Terms—Multi-agent reinforcement learning, Internet of Vehicles (IoV), Resource allocation, Clusters, Base stations, Proximal Policy Optimization (PPO).

I. INTRODUCTION

The Internet of Vehicles (IoV) represents the next era of advanced transportation systems, which can potentially mitigate the issues of conventional transportation systems. IoV provides the foundation for intelligent transportation systems (ITS) by integrating vehicles with built-in advanced communication capability to the Internet. These vehicles interact with each other (V2V) [1], vehicles with pedestrians (V2P) [2], and vehicles with infrastructure (V2I) [3], thus creating a comprehensive Vehicle-to-Everything (V2X) communication network. Advanced transportation applications require prompt response, high data rates, low latency, and real-time communication [4]. Unfortunately, current IoV implementations often need to

meet stringent requirements (e.g., limited data processing capabilities caused by network latency) for optimal performance despite rapid advances in communication technologies [5].

A recent study [6] introduced the management of cells in the mobile network in clusters, where a base station is responsible for the allocation of resources within its coverage area. A hierarchical structure was proposed to efficiently handle geographically distributed clouds by balancing multiple conflicting objectives like availability, reliability, cost, response time, and network traffic. However, the complexity of managing and optimizing resources in geographically distributed clouds remains high due to the NP-hard nature of the problem. More specifically, IoV communication faces specific challenges, such as user location, mobility, dynamic data demand, and unpredictable channel conditions, which increase the complexity of efficiently managing and optimizing resources. Reinforcement Learning (RL) can address the above challenges by making intelligent decisions in dynamic and complex scenarios.

RL can potentially make dynamic and data-driven decisions in dynamic and complex environments. Among the variants of RL algorithms, PPO is suitable for optimizing resource allocation by continually fitting to unstable network conditions and dynamic user demands. Some works, such as Tuong et al. [7], leveraged deep RL to optimize resource management in dense IoV networks. Liang et al. [8] utilized RL to improve network performance by increasing resource allocation. In [9], resource optimization has been carried out with the help of DQN learning, while in [10], DQN learning is used to improve system security. However, these approaches do not cope with

TABLE I: Abbreviations and their definitions

No.	Abbreviation	Meaning
1	MARL	Multi-Agent Reinforcement Learning
2	V2I	Vehicle-to-Infrastructure
3	V2V	Vehicle-to-Vehicle
4	V2P	Vehicle-to-Pedestrian
5	ITS	Intelligent Transportation Systems
6	PPO	Proximal Policy Optimization
7	SAC	Soft Actor-Critic
8	A2C	Advantage Actor-Critic
9	RL	Reinforcement Learning
10	DQN	Deep Q-Network

resource allocation, scalability, and adaptability within cell clusters and do not ensure seamless network-wide integration.

To address the above IoV problems, we introduce a Multi-Agent Reinforcement Learning (MARL) framework based on PPO [11], where each base station within a cluster functions as an independent agent by collaboratively managing transmission power and bandwidth allocation. To validate the effectiveness of this approach, we conducted extensive simulations that compare the performance of the multi-agent PPO [12] with other reinforcement learning algorithms, such as soft actor-critic (SAC) [13] and advantage actor-critic (A2C) [14].

Our main contributions are as follows:

- We present a novel multi-agent reinforcement learning framework utilizing PPO to optimize resource allocation dynamically in clustered IoV networks, specifically targeting high data rate applications.
- Our framework empowers base stations to function as independent, collaborative agents, enabling decentralized management of transmission power and bandwidth, which significantly enhances overall network efficiency.
- The proposed framework is designed for scalability and adaptability, making it suitable for various IoV network sizes and configurations, particularly those with stringent data rate demands.
- Simulations demonstrate that our PPO-based approach consistently outperforms leading reinforcement learning algorithms like SAC and A2C in data rates, bandwidth utilization, and power efficiency, validating its effectiveness in real-time resource management.

The remainder of the paper is structured as follows. Related work is discussed in Section II. Section III provides a detailed description of our system model, including the processes at each level of the cell clusters. Section IV discusses the results of our study, which emphasize the performance improvements achieved through our proposed framework. Performance evaluation and outcomes are discussed in Section VIII. Finally, we conclude this work in Section XI. For the comfort of readers, Table I defines abbreviations used in this paper.

II. RELATED WORK

Several conventional optimization techniques have been extensively utilized to address the challenges of resource allocation in VEC. For instance, [15] introduced a task scheduling

approach that utilizes an enhanced evolutionary algorithm to address the task offloading issue in an end-edge-cloud architecture. This approach considers vehicles' varying computational capabilities and the interconnection between tasks. To address effective offloading of collaborative computations in VEC, [16] put up an optimization strategy based on the alternating direction multiplier approach. To optimize the projected utility of the base station, [17] combined contract theory with matching theory to offer a two-stage resource allocation and task offloading approach.

Deep reinforcement learning (DRL) has emerged as a key method for addressing compute offloading challenges, leading to the development of numerous DRL-based systems. For example, [18] proposed a scheduling technique using a PPO approach to reduce task latency and energy consumption. Similarly, [19] introduced a method addressing offloading and relocation decisions, utilizing the asynchronous dominant actor-critic algorithm to minimize task execution delays and system energy usage. Additionally, [20] presented a reinforcement learning technique called Deep Deterministic Policy Gradient (DDPG), aimed at optimizing offloading, caching decisions, and resource allocation in Vehicular Edge Computing (VEC).

In contrast, [21] proposed a method that combines Lyapunov optimization with deep Q-networks to address the offloading and resource allocation problem in Vehicular networks. The approach aims to maximize task throughput over continuous time periods while accounting for various constraints. However, these centralized scheduling methods require a central agent to collect system status information, and as the number of vehicles increases significantly, the complexity and communication overhead involved in centralizing job offloading decisions also rise sharply. However, we discussed the summary of the above literature, including its methods and shortcomings, in Table II.

III. SYSTEM ARCHITECTURE

A. Clustered Network Framework

The proposed system architecture structures the IoV network into clusters of cells, each managed by a base station responsible for a designated geographical area as shown in Fig. 1. Each base station M_i covers a radius R_i , ensuring broad coverage and efficient high traffic management. The total network coverage $\mathcal{A}$ is represented as the union of the areas covered by all base stations as

$$\mathcal{A} = \bigcup_{i=1}^N \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x} - \mathbf{c}_i\| \leq R_i\}, \quad (1)$$

where $\mathbf{c}_i$ denotes the location of the i -th base station, and N is the total number of base stations. User devices U_j , such as mobile and IoT devices, dynamically connect to the nearest base station M_{nearest} based on their spatial coordinates $\mathbf{u}_j$, given as

$$M_{\text{nearest}} = \arg \min_i \|\mathbf{u}_j - \mathbf{c}_i\|. \quad (2)$$

TABLE II: Summary of Related Works

Reference	Method	Limitations
[15]	Enhanced evolutionary algorithm for task scheduling	May struggle with scalability as vehicle numbers increase, leading to potential delays in decision-making.
[16]	Alternating direction multiplier approach for collaborative computations	Complexity increases with the number of vehicles; may require significant computational resources.
[17]	Combination of contract theory and matching theory for resource allocation	Potentially limited by the assumptions in contract and matching theories, which may not hold in dynamic scenarios.
[18]	PPO-based scheduling technique	Requires extensive training data, which may not be readily available in real-time scenarios.
[19]	Asynchronous dominant actor-critic algorithm for offloading decisions	Higher computational overhead and complexity can lead to longer response times in dynamic environments.
[20]	Deep Deterministic Policy Gradient (DDPG) for offloading and caching	Sensitive to hyperparameter tuning; improper settings can lead to suboptimal performance.
[21]	Lyapunov optimization combined with deep Q-networks	Centralized scheduling increases complexity and communication overhead as the number of vehicles grows.
[Proposed work]	Multi-agent reinforcement learning using PPO	Addressed the above discussed limitations.

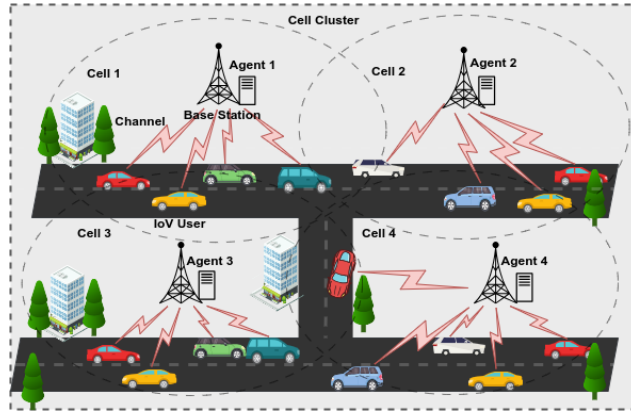


Fig. 1: System model.

This dynamic association optimizes resource allocation across the network, allowing for efficient load balancing and seamless connectivity as users move through different coverage areas.

B. Communication and Channel Assignment Model

The communication model is designed for optimal resource allocation across the network, with each base station dynamically managing its available channels C_i for the connected users U_j . The bandwidth B_{ij} allocated to a user U_j by base station M_i is a function of the user's data rate D_j . The available bandwidth $B_{\max}^i$ is given as

$$B_{ij} = \frac{D_j \cdot B_{\max}^i}{\sum_{k \in U_i} D_k}, \quad (3)$$

where U_i represents the set of users associated to base station M_i . The channel assignment C_{ij} for user U_j is dynamically adjusted based on the signal-to-interference-plus-noise ratio (SINR) γ_{ij} to maximize the quality of service (QoS) as

$$C_{ij} = \arg \max_{c \in C_i} \gamma_{ij}(c), \quad (4)$$

where $\gamma_{ij}(c)$ is the SINR for user U_j on channel c , computed as

$$\gamma_{ij}(c) = \frac{P_i \cdot h_{ij}}{\sigma^2 + \sum_{k \neq i} P_k \cdot h_{kj}}, \quad (5)$$

where, P_i is the transmission power of base station M_i , h_{ij} is the channel gain between M_i and U_j , and σ^2 is the noise power.

C. Mobility and Connectivity Model

The mobility model simulates the dynamic movement of users within the network. It continuously assesses users' proximity to different base stations. As users move, the system adjusts the channel assignments and resource allocations in real-time to maintain optimal connectivity and service quality. The optimal connectivity for a user U_j to a base station M_i is determined by maximizing the SINR γ_{ij} , as shown in (4). This ensures that each user is connected to a base station that provides the highest quality signal, thereby minimizing interference and enhancing overall network performance.

To ensure seamless handover as users transit between the coverage areas of different cells, the system evaluates the

signal strength S_i from the current base station. It compares S_i with the interference I_i from neighboring stations as

$$C_{\text{handover}} = \arg \max_{i \in \{M_i, M_k\}} (S_i - I_i) \quad (6)$$

This relation in (6) governs the handover process, ensuring that users maintain consistent connectivity with minimal service degradation as they move through the network.

IV. OPTIMIZATION OBJECTIVES

This research aims to enhance the distribution of transmission power (P_i) and bandwidth (B_{ij}) in a clustered Internet of Vehicles (IoV) network, with the goal of maximizing the aggregate data rate (r_i) for users, while ensuring effective resource utilization and reducing interference. The data rate r_{ij} for user U_j serviced by base station M_i is expressed as:

$$r_{ij} = B_{ij} \cdot \log_2 \left(1 + \frac{P_i \cdot h_{ij}}{\sigma^2 + I_{ij}} \right), \quad (7)$$

where h_{ij} is the channel gain between M_i and U_j , σ^2 is the noise power, I_{ij} represents the interference experienced by U_j from other base stations, the optimization problem can be formulated as:

$$\max_{\{P_i, B_{ij}\}} \sum_{i=1}^N \sum_{j=1}^{U_i} r_{ij} \quad (8)$$

Subject to:

$$C_1 : \sum_{j=1}^{U_i} B_{ij} \leq B_{\max}^i, \quad (9)$$

$$C_2 : 0 \leq P_i \leq P_{\max}^i \quad (10)$$

where C_1 and C_2 denote constraints, $B_{\max}^i$ is the maximum bandwidth available at base station M_i , and $P_{\max}^i$ is the maximum transmission power for M_i .

The goal is to optimize the aggregate data rates throughout the network by judiciously distributing bandwidth and power, while ensuring that the total bandwidth utilized does not above the available thresholds and that power levels stay within safe operational limits. The framework seeks to optimize user engagement while ensuring superior service quality.

V. PROPOSED MULTI-AGENT FRAMEWORK

This section introduces an advanced multi-agent reinforcement learning (MARL) architecture, illustrated in Fig. 2, specifically engineered to optimize transmission power and bandwidth distribution in clusters of base stations inside Internet of Vehicles (IoV) networks utilizing the Proximal Policy Optimization (PPO) method. The PPO agents are designed to enhance overall network performance by evaluating aspects such as user location, mobility, data demand, and channel conditions, thereby ensuring effective resource usage in dynamic circumstances. The architecture utilizes the MARL paradigm, enabling each base station to function as an autonomous agent, hence facilitating decentralized and collaborative decision-making to improve overall network performance.

A. State

We set the state space S_t^i for each agent (base station M_i) at time t , which is designed to get the essential information for optimizing resource allocation within the IoV environment. It is represented as:

$$S_t^i = \{\mathbf{u}_j, D_j, h_{ij}, \gamma_{ij}, P_i, B_{ij}, U_i, \mathcal{N}_i\}, \quad (11)$$

In this context, $\mathbf{u}_j$ signifies the position of the user U_j in relation to base station M_i . At the same time, D_j denotes the data demand of U_j , illustrating the user's fluctuating need for data transmission. The term h_{ij} indicates the channel gain between base station M_i and user U_j , and γ_{ij} represents the SINR for U_j , reflecting the quality of the communication link. Additionally, P_i indicates the transmission power of base station M_i , and B_{ij} refers to the bandwidth allocated to U_j . The set U_i comprises all users now connected to base station M_i , whereas $\mathcal{N}_i$ denotes the adjacent base stations, whose impact may influence resource management due to interference and collaboration necessities.

B. Action

The action space A_t^i for each agent is formulated to enable dynamic resource management by adjusting key network parameters. It includes the following continuous actions:

$$A_t^i = \{\Delta P_i, \Delta B_{ij}\}, \quad (12)$$

where ΔP_i represents the adjustment in the transmission power of base station M_i , allowing the agent to optimize power allocation while controlling interference and ensuring energy efficiency, and ΔB_{ij} denotes the adjustment in bandwidth allocated to the user U_j , enabling the agent to distribute bandwidth based on user demand and network conditions adaptively. These actions collectively aim to maximize overall network performance by dynamically optimizing power and bandwidth allocation, ensuring efficient resource utilization while minimizing interference across the network.

C. Reward

The reward function R_t^i is designed to guide each agent towards actions that improve network performance by considering multiple factors:

$$R_t^i = \alpha \cdot \sum_{j=1}^{U_i} r_{ij} - \beta \cdot P_i + \gamma \cdot B_i, \quad (13)$$

where r_{ij} represents the data rate for user U_j served by base station M_i , P_i is the transmission power, which is penalized for encouraging power efficiency, and B_i represents the efficiency of dynamic bandwidth assignment to the base station based on the current load or the total users' data rate. The coefficients α , β , and γ balance the trade-offs among data rate maximization, power efficiency, and dynamic bandwidth allocation efficiency.

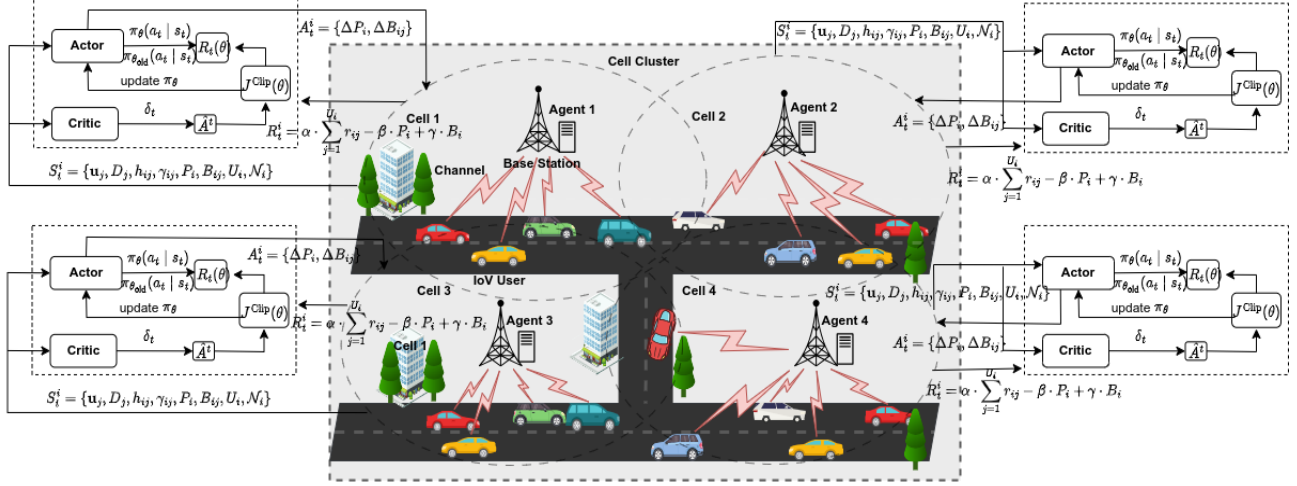


Fig. 2: Proposed Multi-Agent Framework

D. Learning Process

The PPO agents operate in a continuous learning loop. Each agent interacts with the environment (the IoV network) by taking actions based on its policy $\pi_{\theta}(a_t | s_t)$. After each iteration, it receives rewards and updates its policy through on-policy updates that incorporate a clipped surrogate objective. The goal is to learn a policy that maximizes the expected cumulative reward over time, ensuring that the network adapts to real-time changes and maintains optimal performance under varying conditions.

$$\theta_{t+1} \leftarrow \theta_t + \eta \nabla_{\theta} \mathbb{E} \left[R_t^i + \gamma \mathbb{E}_{s_{t+1}^i} V_{\phi}(s_{t+1}^i) \right], \quad (14)$$

where η is the learning rate and $V_{\phi}(s_{t+1}^i)$ is the value function estimating the expected future rewards.

By leveraging MARL with PPO, the framework offers a scalable, decentralized solution for resource management in IoV networks, addressing the challenges posed by high data rate demands and dynamic network conditions.

VI. COMPLEXITY OF THE PROPOSED METHOD COMPARED TO EXISTING APPROACHES

The computational complexity of the proposed Multi-Agent Mobile Network optimization method using Proximal Policy Optimization (PPO) can be compared with traditional reinforcement learning algorithms like Soft Actor-Critic (SAC) and Advantage Actor-Critic (A2C). In this environment, PPO offers a balance between computational efficiency and performance. Each base station operates as an independent agent managing resource allocation, such as bandwidth and transmission power, across users.

The complexity of PPO per iteration is approximately $\mathcal{O}(N \times U \times A \times K)$, where N is the number of agents (base stations), U is the number of users, A is the action dimension, and K is the number of policy updates. SAC due to its dual-network structure, has a higher complexity

Algorithm 1 Proposed Multi-Agent Framework for IoV Networks

- 1: Initialize environment $\mathcal{E}$ with N base stations and M users.
- 2: Initialize PPO agents π_{θ^i} for each base station M_i , with initial policy parameters θ^i .
- 3: **for** each episode **do**
- 4: Reset environment $\mathcal{E}$ to obtain initial states S_0^i for all agents M_i .
- 5: **for** each time step t within the episode **do**
- 6: **for** each agent M_i **do**
- 7: Observe current state S_t^i .
- 8: Select action $A_t^i = \{\Delta P_i, \Delta B_{ij}\}$ based on policy $\pi_{\theta^i}(A_t^i | S_t^i)$.
- 9: Execute action A_t^i in the environment.
- 10: **end for**
- 11: Obtain rewards R_t^i and next states S_{t+1}^i for each agent.
- 12: Store transitions $(S_t^i, A_t^i, R_t^i, S_{t+1}^i)$ in the replay buffer.
- 13: Compute the reward for each agent M_i :
- $$R_t^i = \alpha \cdot \sum_{j=1}^{U_i} r_{ij} - \beta \cdot P_i + \gamma \cdot B_i$$
- 14: Update policy parameters θ^i using the PPO algorithm:
- $$\theta_{t+1}^i \leftarrow \theta_t^i + \eta \nabla_{\theta^i} \mathbb{E} \left[R_t^i + \gamma \mathbb{E}_{s_{t+1}^i} V_{\phi}(s_{t+1}^i) \right]$$
- 15: **if** episode terminates **then**
- 16: Break the time step loop.
- 17: **end if**
- 18: **end for**
- 19: Log performance metrics such as total reward, data rate, bandwidth usage, power consumption, and SINR.
- 20: **end for**

of $\mathcal{O}(N \times U \times A^2 \times G)$, making it more computationally expensive for large-scale networks. A2C, on the other hand, has a simpler complexity of $\mathcal{O}(N \times U \times A)$ but struggles in dynamic environments due to its synchronous updates. Thus, PPO provides a scalable solution with moderate complexity, making it well-suited for real-time resource allocation in clustered IoV networks.

VII. CHALLENGES AND SOLUTIONS IN MULTI AGENT PPO IMPLEMENTATION

Optimization of resource utilization for the IoV environment through the PPO algorithm faces many challenges. One of

the first challenges was to define an appropriate state and action space, being that it should encapsulate dynamic network conditions such as user locations, data rates, and channel states but also allow us to mitigate wireless transmission power and bandwidth finely. To accomplish this, we needed to include many network parameters in the state space with a continuous action space that also accounted for smooth adaptive control. Finally, the action space was large due to the considerable number of base stations and users, which posed another severe challenge for exploration with a slow convergence speed. We addressed this by normalizing the action space accordingly and tweaking the reward function so that it would discourage excessive resource usage (as much as possible/learning can be tricked with false signal) in accordance with using PPO, which has a continuous action space.

Further challenges are introduced by rewards being delayed, making it challenging for agents to learn optimal policies in a timely manner and needing to trade power efficiency between maximization of data rate. This required designing a reward function that reflected both instantaneous and long-term value in terms of network performance metrics, as the challenge was to ensure agents learned to strike a balance between short-term rewards and long-term efficiency. Scalability was another bottleneck, with a large number of agents in a centralized framework having to make independent decisions; therefore, we adopted a decentralized, multi-agent approach in which all base stations independently made decisions but cooperatively. It provided a way in which the system could scale elegantly while allowing for efficient use of resources throughout.

VIII. PERFORMANCE EVALUATION

A. Simulation Setting and Experiment

In this section, the performance of the proposed Multi-Agent Reinforcement Learning (MARL) framework is evaluated in a simulation environment that represents the clustered mobile network surroundings. The simulation environment covered four base stations and simulated 100 users associated with different mobility patterns. Various speeds is assigned to simulate both pedestrian and vehicular modes of movement, providing a rich mobility scenario for its resource allocation capabilities.

A custom Gym environment in Python is created to develop the simulation environment that correctly models base stations, users and communication channels. To capture the heterogeneity of real networks, base stations were initialized with different transmission powers and coverage areas. Users moved stochastically all across the simulation space, and we continuously changed their mobility in order to simulate real-world dynamics.

The first round of evaluations is concerned with how the network could perform under mobile conditions, mainly managing users that move across different data rate requirements. The network load is then gradually increased throughout several simulations by using more active users and demanding a higher data rate from each user. It helped to gain precious

knowledge about the network performance capability to sustain peak load conditions, resource distribution management and service quality.

One important aspect we focused on is seamless handoffs between base stations, ensuring that users could move across the coverage zones without degradation of service. Whether the system was able to handle these handoffs with minimum disruption is necessary as a metric for overall performance since uninterrupted service is one of the main necessities in mobile networks.

The simulation methodology is comprehensive, encompassing a diverse range of scenarios from controlled settings to more dynamic and stress-inducing conditions. This thorough testing methodology confirmed that the suggested framework could satisfy theoretical performance criteria and adeptly respond to practical obstacles. Comprehensive simulation parameters are included in Table III.

B. Numerical Results and Analysis

The MARL framework's performance was extensively evaluated using simulations that employed the Stable Baselines3 and PettingZoo libraries. Although the primary focus was on the PPO algorithm, comparative assessments were conducted against Soft Actor-Critic (SAC) and Advantage Actor-Critic (A2C) algorithms to ensure a thorough evaluation across multiple dimensions.

Key performance metrics evaluated included power consumption, bandwidth utilization, data rate, and user association. Power consumption was measured based on the transmission power utilized by each base station, while bandwidth utilization assessed the efficiency of allocating and using available bandwidth across the network. The data rate per user was aggregated to reflect overall network throughput, and user association metrics tracked the number of users successfully connected to base stations, providing insights into coverage and service quality.

The experimental design and the MARL framework's implementation facilitated detailed analysis and comparison, allowing us to observe how each algorithm performed under varying conditions. The results demonstrated that the MARL framework, particularly when using PPO, consistently outperformed SAC and A2C in terms of achieving higher data rates, optimizing bandwidth utilization, and maintaining efficient power consumption.

TABLE III: Simulation Parameters

Parameter	Value
Number of Base Stations or Cells	4
Total Number of Users	100
User Speed Range (km/h)	0 to 50
Frequency Band	2.5 GHz to 3.7 GHz
Height of Stations (meters)	20
Receiver Height (meters)	1.5
Transmission Power of Stations (mW)	10,000
Area Size (square meters)	5,000 x 5,000
Coverage Area Calculation	Path Loss Model

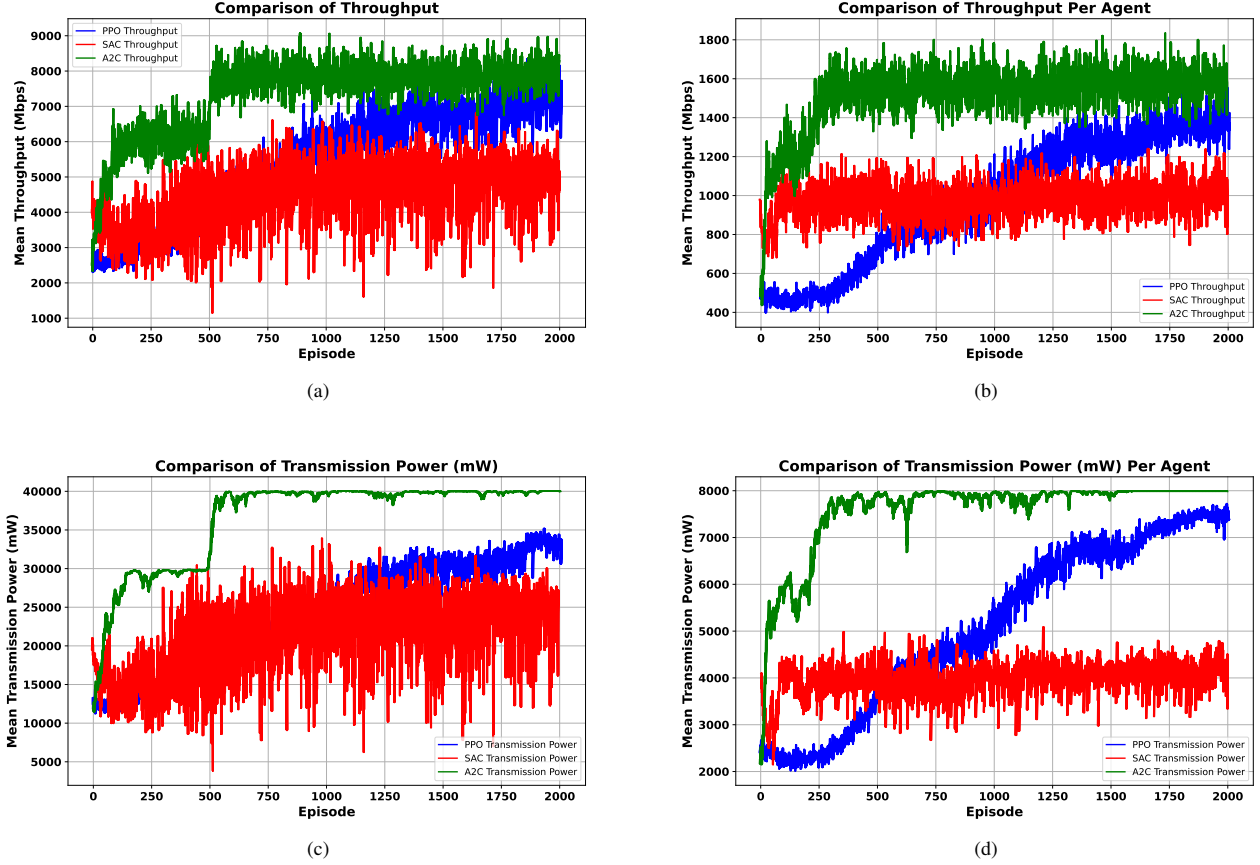


Fig. 3: a) Comparison of the total mean throughput. b) Comparison of throughput per agent. c) Comparison of total transmission power. d) transmission power per agent.

1) *Data Rate Comparison*: In Fig. 3 (a) and (b), the multi-agent SAC algorithm achieves the highest total data rate among the three algorithms compared. The graph shows that the mean SAC throughput increases steadily as the number of episodes increases. The PPO algorithm performs well, but with a slightly lower total data rate than SAC. The A2C algorithm has a lower total data rate than SAC and PPO. The mean throughput for A2C gradually increases with increasing episodes. In summary, the multi-agent SAC algorithm outperforms the other baseline algorithms in terms of data rate due to the management of resources in a clustered environment.

2) *Transmission Power Comparison*: Fig. 3 (c) and (d) shows that the multi-agent SAC algorithm has the most balanced power consumption compared to the PPO and A2C algorithms. The A2C algorithm has the highest total transmitting power among the three algorithms across the given number of episodes. SAC's performance demonstrates its efficiency in managing power consumption while achieving high data rates, making it a more power-efficient choice than the other algorithms.

3) *Bandwidth Utilization Comparison*: According to Fig. 4 (a) and (b), the multi-agent SAC algorithm optimally manages total bandwidth and total throughput, performing better than PPO and A2C. The results indicate that SAC effectively allocates bandwidth while meeting the increasingly dynamic data rate demands, highlighting its capability to manage network resources efficiently under varying conditions.

4) *User Association Comparison per Agent*: As shown in Fig. 4 (c) and (d), the multi-agent SAC algorithm covers the highest number of users in base stations compared to PPO and A2C. The ability of SAC to dynamically adapt to real-time network conditions and efficiently manage resources ensures that a larger number of users are served, maintaining high-quality service across the network. The results show that the multi-agent PPO consistently outperforms the other algorithms in data rates, power efficiency, and overall network performance, highlighting its effectiveness in optimizing resource allocation within clustered IoV networks.

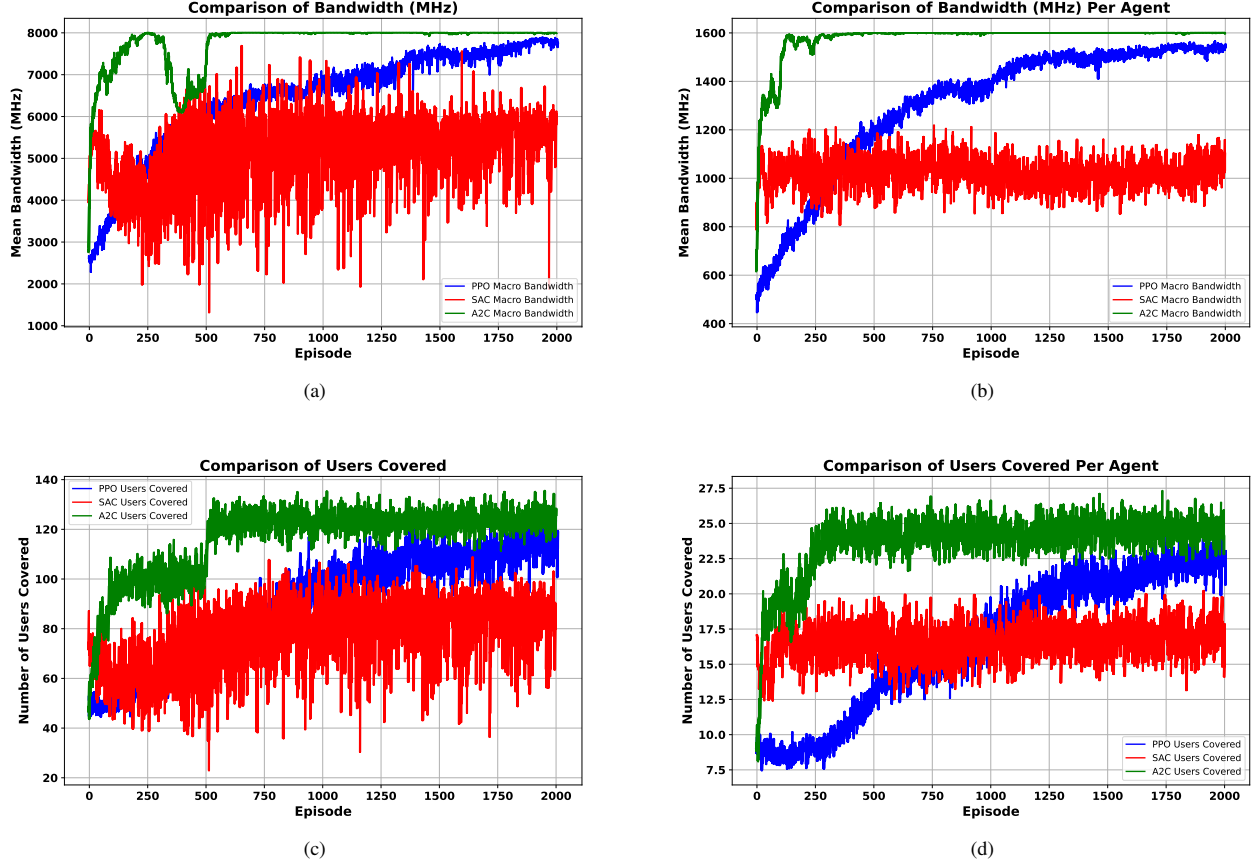


Fig. 4: a) Comparison of mean bandwidth allocation. b) Comparison of bandwidth allocation per agent. c) comparison of mean for user served. d) Comparison of per agent user served.

IX. A NOTE ON THE SCALABILITY OF THE PROPOSED MECHANISM

The proposed MARL framework based on PPO offers a scalable solution for resource allocation in the IoV environments. By enabling each base station to act as an independent agent within a cluster, the framework can efficiently adapt to variations in network size and density. This decentralized approach ensures that as the number of vehicles, base stations, or communication nodes increases, the system can maintain its performance by distributing the computational load among agents. Furthermore, the collaborative nature of the agents allows the framework to seamlessly integrate into larger networks, ensuring consistent management of resources such as transmission power and bandwidth, even in dense IoV settings. This scalability makes the approach well-suited for real-world deployment where network conditions and demands are continuously evolving.

X. FUTURE WORKS

The suggested MARL paradigm demonstrates encouraging outcomes. However, other avenues for future research exist.

One approach is to incorporate the framework with 5G and 6G technology. This could further augment its capacity. It would satisfy ultra-low latency and high throughput criteria. A potential area of emphasis may involve more intricate mobility models. Integrating real-world traffic patterns is crucial to representing the dynamic characteristics of vehicular settings accurately.

The framework is modified to accommodate various network situations, including diverse communication standards and geographical challenges. These enhancements augment flexibility. Ultimately, transfer learning may be investigated. It may facilitate agents' rapid adaptation to novel settings. This would decrease training duration. This would enhance the framework's practicality across many IoV scenarios.

XI. CONCLUSION

This research presents a multi-agent reinforcement learning framework employing the SAC method to enhance resource allocation in clustered IoV networks. Simulation findings demonstrate that the framework effectively manages transmission power and bandwidth dynamically, markedly improving

network performance. The SAC algorithm surpasses PPO and A2C for data rate, power efficiency, bandwidth utilization, and user association. A multi-agent SAC is identified as a resilient option for meeting the intricate requirements of ITS. Subsequent research will investigate the framework's use in practical contexts to assess its scalability and efficacy.

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